

What we are building, and why

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This document assumes you know nothing about air quality modelling. Every idea is explained twice: once in ordinary language, and once in the technical form you would need to defend it. Read the green boxes to understand the project. Read the blue boxes when you want the detail underneath.

IN PLAIN WORDS

Delhi's winter smog is so thick it blocks sunlight. Less sunlight means the ground stays cold. Cold ground means the dirty air cannot rise and escape. So the smog gets thicker, which blocks more sunlight. It feeds itself.

Almost every air quality forecast ignores that circle. We built the circle.

THE TECHNICAL VERSION

Aerosol radiative forcing suppresses the surface sensible heat flux, which reduces convective boundary layer growth and the turbulent momentum transport that drives surface wind. The resulting reduction in ventilation raises surface concentration, which raises aerosol optical depth. Because the final term feeds the first, the system must be solved rather than evaluated.

How to read this: sections 1 to 3 are the "what and why", and everyone should read them. Sections 4 to 7 are how it works. Section 8 is what is not finished. Section 9 is a glossary: if a word confuses you anywhere in this document, it is defined there.

1 The problem we were given

1.1 What the Ministry actually asked for

The Ministry of Earth Sciences ran a problem statement called *Air Pollution and Weather Coupled Forecasting System, Delhi NCR focus*. Stripped of jargon, it asks for four things.

THEY ASKED FOR	WHICH MEANS
A forecast that links weather and pollution in both directions	Not just "wind blows pollution away", but also "pollution changes the weather". This is the hard half and the whole point.
Special attention to temperature inversions	The invisible lid that forms over Delhi on winter nights and traps everything underneath it.
Handling of stubble burning smoke	Farmers in Punjab and Haryana burn crop residue. That smoke travels 200 to 400 km and arrives over Delhi. The forecast has to know when it will actually land.
A dashboard with a 72-hour outlook	Three days ahead, at high enough resolution to be useful inside the city, that a non-specialist can operate.

1.2 Why the "both directions" part is the whole challenge

IN PLAIN WORDS

Think of a pot with a lid. The city is the pot. Cars, factories and fires keep adding smoke. On a normal day the lid is high, so the smoke spreads through a lot of air and stays thin. On a winter night the lid drops very low, so the same smoke is packed into a tenth as much air. Nothing about the smoke changed. The pot got smaller.

Now the part that most forecasts miss. That thick smoke blocks the sun. The ground does not warm up. When the ground does not warm up, the lid stays low. So tomorrow the smoke is packed into a small pot again, and it gets thicker still.

THE TECHNICAL VERSION

The mixed layer depth sets the dilution volume for surface emissions. Nocturnal radiative cooling produces a surface based temperature inversion, collapsing that depth from roughly 2,000 m to under 200 m and raising concentration by an order of magnitude at constant emission.

The feedback runs through the radiation budget: column aerosol attenuates incoming shortwave, the reduced surface energy budget lowers the sensible heat flux, and encroachment growth of the convective boundary layer weakens. Published Delhi values put the shortwave reduction at 5 to 35 percent and the boundary layer suppression at 5 to 35 percent under haze.

The sentence to remember

Weather decides how much air the pollution gets to spread into, and the pollution decides how much air the weather will give it tomorrow. That circle is called a feedback loop, and closing it is what makes this project different from a normal AQI app.

2 What we built, in one page

IN PLAIN WORDS

We built a website that predicts Delhi's air quality for the next three days, across about a thousand small squares on the map, and shows you *why* it thinks so rather than just giving you a number.

2.1 The journey of one forecast

Here is what happens, in order, every six hours.

STEP	WHAT HAPPENS	IN PLAIN WORDS
1	Collect	We download four things: readings from 159 real air quality monitors around Delhi, a weather forecast, a global pollution model's output, and satellite photos of where fires are burning.
2	Compute the physics	From the weather we work out things the weather forecast does not directly tell us: is there a lid tonight, how high is it, how much air is available for mixing, how much sunlight is getting through the haze.
3	Track the smoke	Every fire the satellite saw becomes a puff of smoke that we push along on the forecast wind, hour by hour, to see if and when it arrives over Delhi.
4	Ask the AI	Twelve trained models take all of the above and predict four pollutants at three different times ahead.
5	Close the loop	We take that prediction and run the feedback circle: this much smoke blocks this much sun, which lowers the lid this much, which raises the smoke this much. Repeat until the answer stops changing.
6	Publish	Convert to the official Indian AQI number, work out the chance of crossing each emergency threshold, and put it on the website.

2.2 What the website shows

PAGE	THE QUESTION IT ANSWERS
What this is	What the project is and how it meets the brief.
Where the air comes from	The map, and where today's fires are burning.
The lid over the city	Is there an inversion tonight, how low, and how long.
The loop we close	The feedback circle, with each step's live number.
Smoke from the fields	Whether the stubble smoke actually reaches us.
The 72-hour outlook	The actual forecast: AQI per square, and emergency risk.
Does it actually work	Every accuracy number, including the ones that went against us.

3 The AI: what it is and what it does

This is the section most people want and most people find confusing, so it goes slowly.

3.1 How many models, and what is each one for?

IN PLAIN WORDS

We have **twelve** separate prediction models, plus five extra ones for measuring uncertainty. Seventeen in total.

Why twelve? Because we predict **four different pollutants** at **three different times ahead**. Four times three is twelve. Each model does exactly one job: for example, "predict PM2.5, 24 hours from now". A different model predicts ozone 72 hours from now. They do not share anything.

THE TECHNICAL VERSION

Twelve independent gradient boosted regression models, one per (target, horizon) pair over PM2.5, PM10, NO₂ and O₃ at 24, 48 and 72 hours, plus five quantile heads at the 10th, 25th, 50th, 75th and 90th percentiles for PM2.5 at 24 hours. Separate models rather than a multi-output model because the targets have materially different drivers and seasonality.

POLLUTANT	WHAT IT IS, IN PLAIN WORDS	24 H	48 H	72 H
PM2.5	Tiny soot particles. The dangerous one. Gets into your lungs and bloodstream.	model 1	model 2	model 3
PM10	Bigger dust particles. Road dust, construction, desert dust.	model 4	model 5	model 6
NO ₂	A gas from burning fuel. Mostly vehicles.	model 7	model 8	model 9
O ₃ (ozone)	Not emitted directly. It is <i>made</i> in the air by sunlight acting on pollution. Peaks on bright afternoons.	model 10	model 11	model 12

Ozone is the odd one out and it matters: because sunlight makes it, anything that blocks sunlight (like smog) reduces it. That is why our haze calculations affect ozone in the opposite direction to the others.

3.2 What actually goes into a model, and what comes out

IN PLAIN WORDS

Think of one model as a very experienced clerk. Every hour you hand the clerk a form with **142 numbers** on it: today's pollution readings, yesterday's, the temperature, the wind, whether there is a lid, how much sunlight is getting through, what month it is, whether it is Diwali. The clerk hands back **one number**: what PM2.5 will be 24 hours from now.

The clerk learned the job by being shown **571,037 real examples** from the past six years: here is what the form looked like, and here is what actually happened a day later. Do that half a million times and the clerk gets good at it.

THE TECHNICAL VERSION

142 features per row. The supervised frame is built by shifting the meteorological columns backward by the horizon, so at time t the model sees forecast meteorology valid for $t+h$, exactly as an operational system would. Pollution columns are never shifted backward, which prevents target leakage.

Training set 571,037 station hours over 44 CPCB stations, Oct 2020 to Aug 2026, with 1 Nov 2025 to 28 Feb 2026 withheld entirely.

3.3 What kind of AI is it? (Not a chatbot, not a neural network)

IN PLAIN WORDS

It is a **decision tree** method. A decision tree is a flowchart of yes or no questions: "is the wind under 2 m/s? yes. Is there a lid? yes. Is it December? yes. Then predict high pollution."

One tree is weak. So the method builds hundreds of them in a row, where each new tree specifically studies the mistakes the previous trees made and tries to correct them. That is what "gradient boosting" means: a committee of simple flowcharts, each one patching the last one's errors.

We did not use a neural network. For table shaped data of this size, boosted trees usually win, they cope with missing readings, and crucially **we can see which inputs they relied on**, which is how we caught a real bug once.

THE TECHNICAL VERSION

XGBoost, `reg:squarederror`, `max_depth 6`, `eta 0.04`, `subsample 0.8`, `colsample_bytree 0.8`, `min_child_weight 8`, `reg_lambda 1.5`, up to 700 rounds with early stopping at 40. Depth is deliberately shallow: deeper trees mainly memorised individual episodes rather than learning regime behaviour.

3.4 What did it actually learn?

We can rank which inputs the model leaned on. For PM2.5 at 24 hours, in order: the day of the year, whether it is winter, the day of year again in a cyclical form, whether it is monsoon, the reading one hour ago, the average over the last 72 hours, and then **the ventilation coefficient**.

WHY THAT LAST ONE MATTERS TO US

The ventilation coefficient is not something the weather service gives you. It is something **we calculate**: how tall the mixing layer is, multiplied by how fast the wind is blowing through it. In other words, how much air per second is available to carry pollution away.

The model ranked our calculated number above most of the raw weather data it was made from. That is the physics earning its place, and it is a good thing to be able to say out loud.

4 The physics: the loop, in detail

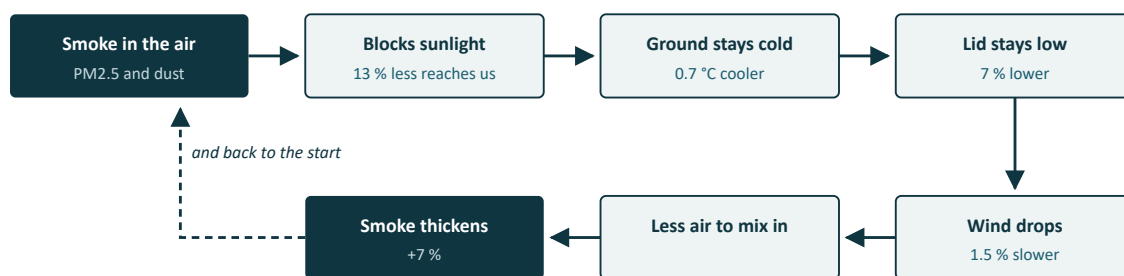


Figure 1. Percentages are real measured values from the system running now, during hazy daylight hours.

WHY THIS IS HARD, AND WHY IT IS CALLED "SOLVING"

Look at the dashed arrow. The last box feeds back into the first one. So you cannot just calculate down the chain once and stop, because the answer you get at the end changes the number you started with.

What we do instead is go round the circle repeatedly. Calculate, feed the answer back in, calculate again. After about five laps the number stops changing, and that settled value is the answer. That is what "solving" means here, as opposed to just "calculating".

THE TECHNICAL VERSION

Damped fixed point iteration with relaxation factor 0.5. Every response is clipped to a physically bounded range, so divergence is impossible by construction. Any cell that fails to converge within the iteration budget falls back to the uncoupled value and is counted in the diagnostics. Current run: 5 iterations, converged, zero fallbacks.

4.1 How we know these numbers are not invented

IN PLAIN WORDS

Scientists have measured these effects over Delhi and published the results. We compare every number our system produces against those published ranges. If any of them lands outside, the software build **fails**. It is a test, not a report.

Right now all eight land inside. And we never showed the system those published numbers, so it is not copying them.

WHAT CHANGES UNDER HAZE	OURS	PUBLISHED RANGE	RESULT
Sunlight reaching the ground	13.2 % less	5 to 35 %	inside
Daytime temperature	0.72 °C cooler	0.1 to 2.5 °C	inside
Height of the mixing layer	6.9 % lower	5 to 35 %	inside
Wind speed	1.5 % slower	0.5 to 6 %	inside
Resulting PM2.5	7.2 % higher	2 to 30 %	inside
Resulting PM10	5.8 % higher	1 to 25 %	inside
Resulting NO ₂	16.3 % higher	2 to 35 %	inside
Resulting ozone	slightly lower	must go down	correct direction

5 The smoke from the fields

IN PLAIN WORDS

Satellites photograph the earth several times a day and can spot fires by their heat. Every October and November, thousands of them light up across Punjab and Haryana as farmers clear crop stubble.

We take each fire the satellite saw and turn it into an imaginary puff of smoke. We then push each puff along on the forecast wind, hour by hour, letting it spread out as it travels, and see where it ends up.

The important part: arriving over Delhi is not the same as being breathed in Delhi. If the smoke is floating above the lid, it passes over the city and nobody at ground level notices. It only affects people when it gets mixed down to the ground.

THE TECHNICAL VERSION

Lagrangian puff dispersion. Emission mass is derived from fire radiative power per detection, advected on the forecast wind field, with Gaussian horizontal spread as a function of travel time and stability class. Surface coupling is decided by the relationship between puff height and the diagnosed mixing depth.

Something we recently discovered we were doing wrong

We used to decide "does this smoke reach the ground" with a simple height comparison: if the lid is above the smoke, it arrives; otherwise it does not. That is a switch, and the real atmosphere does not have a switch.

What really happens is that overnight the smoke sits in a layer above the city, completely cut off from the ground. Then after sunrise the ground warms, the mixing layer grows upward, and it gradually eats into that smoke layer, pulling it down over about two to three hours. That is why Delhi's pollution has a sharp **morning peak between 7 and 10 am** that traffic alone cannot explain.

We have now built a two-layer version that reproduces that morning peak from a proper mass budget. It is written and tested but not yet switched on by default, because we want to score it against the Ministry's own data first. Promoting it because it is newer would be the wrong reason.

6 How we know it works

6.1 The baseline that matters

IN PLAIN WORDS

Any forecast has to beat the laziest possible forecast, which is **"tomorrow will be the same as today"**. That is called persistence, and it is embarrassingly hard to beat, because air pollution really is quite similar from one day to the next.

If you cannot beat persistence, your model is worthless no matter how sophisticated it looks. All twelve of ours beat it, by between 15 and 29 percent.

6.2 The four tests, and what each one rules out

TEST	WHAT IT PROVES, IN PLAIN WORDS	RESULT
Hide a whole winter	We deleted four months of winter from the training data, trained the models without ever seeing it, then asked them to predict it. This rules out "it only works because it memorised last winter".	12 of 12 beat persistence
Hide a whole station	We deleted one monitoring station completely, then asked the model to predict that exact location. This is the real test, because we forecast 1,120 map squares and only about 40 have a monitor in them.	10 of 10 improved, +31.6 %
Ask an impossible question	"What would ozone do if the smog halved?" That has never happened in the data, so a pure pattern matcher cannot answer. Ours says +12.8 %; published Delhi science says +25 %. Same direction, right ballpark, and we never showed it that number.	same sign, within 2×
Check the honesty of "maybe"	When the system says "80 percent confident", is it right 80 percent of the time? Measured: 75.6 percent. Slightly cautious, which is the safe direction to be wrong in.	well calibrated

6.3 The result that went against us, which we report anyway

IN PLAIN WORDS

We tested whether the feedback loop, the thing the whole project is named after, actually makes the forecast more accurate. **It does not.** Overall it is 0.11 percent worse.

But split it up and it gets interesting: on hazy days it is 0.35 percent *better*, and on clean days it is slightly worse. In other words, it helps exactly where the physics says it should and does nothing where the physics says it should not. That is what a real but weak effect looks like.

We report this openly. A system that only shows you its wins has not been tested, it has been advertised. And what the loop genuinely buys us is not accuracy: it is the ability to answer "what if" questions, and to explain *why* a number is high.

7 What is not finished

7.1 We do not run a full chemistry model, and cannot

IN PLAIN WORDS

There is a well known scientific program called WRF-Chem that simulates weather and chemical reactions together, properly, from first principles. The brief mentions it. We do not run it, and it is not because we did not try.

We measured what it needs. Just *compiling* the program takes up to four hours on the free computers we have, out of a six hour limit. Then actually running a three day forecast takes twelve to eighteen hours. And it needs ten to twelve gigabytes of memory when only sixteen are available with no safety margin. It is a national research centre's programme, with a cluster behind it.

What we do instead is use the output of a model like that which is **already running**: the European CAMS system, which does have online chemistry, and we add our own physics on top.

THE TECHNICAL VERSION

WRF-Chem with MOZART gas phase and MOSAIC aerosol on a $42 \times 33 \times 35$ grid at 10 km requires 4,320 integration steps for 72 hours. Benchmarked at 12 to 18 hours wall clock on 4 cores, with peak RSS of 10 to 12 GB driven by KPP solver intermediates, and no swap on a CI runner. CMAQ and CAMx fail on ephemeral disk during meteorological ingest before the model starts.

7.2 The rest of the list

NOT DONE	WHAT IT MEANS AND WHY IT MATTERS
Chemistry in one layer only	We treat the air over each square as one well mixed box. The real atmosphere has structure with height. We track that structure and draw it, but the chemistry does not use it yet. This is our biggest simplification.
No hydrocarbon chemistry	Ozone formation depends on two families of chemicals. We model the sunlight half properly and do not model the hydrocarbon half at all, because nobody measures hydrocarbons here.
Observations are not assimilated	We use the monitor readings to train and to set the starting point, but not in the formal way weather centres blend observations into a model state.
Two-layer smoke model not switched on	Built and tested, waiting to be scored against the Ministry's data before we promote it.

How to talk about all this

Do not be defensive about any of it. Every one of these is a known, named, measured limitation with a reason. Saying "we measured what WRF-Chem needs and it does not fit, so we consume an operational coupled model instead" is a much stronger answer than hoping nobody asks.

8 If you remember only five things

- 1** **Pollution changes the weather, not just the other way round.**
Smog blocks sunlight, the ground stays cold, the lid stays low, and the smog gets worse. Closing that circle is what the brief asked for and what we built.
- 2** **Twelve AI models. Four pollutants, three time horizons each.**
Each takes 142 numbers about the atmosphere and returns one prediction. They learned from 571,037 real hours of measurements from 44 stations.
- 3** **The AI does not invent the physics. We compute it and feed it in.**
The lid height, the mixing depth, the sunlight getting through the haze: we calculate those and hand them to the model as inputs. Then we apply physics again to its output.
- 4** **Everything is checked against published science we never trained on.**
Eight out of eight of our feedback numbers land inside ranges measured by other scientists over Delhi. If one falls outside, the build fails.
- 5** **We report what did not work.**
The feedback loop does not improve accuracy overall. We say so, on the website and in every document. That is why the numbers that *are* good can be believed.

If someone asks you a question you cannot answer

Say "I do not know that part in detail, but I can tell you where it is written down." Every number in this project comes from a file in the repository, and the full technical report explains each one. Guessing in front of a jury is far worse than saying you will check.

9 Glossary

Every term used in this document or on the dashboard, in plain words.

AQI	Air Quality Index. A single 0 to 500 number combining several pollutants, so the public gets one figure instead of six. India has its own official formula (CPCB) which we follow exactly. <i>0 to 50 is Good, above 400 is Severe.</i>
PM2.5 / PM10	Particulate matter under 2.5 and 10 micrometres across. PM2.5 is the dangerous one because it reaches deep into the lungs. <i>The number is the size, not the amount.</i>
Ozone (O₃)	A gas that is not emitted by anything. Sunlight acts on other pollution and creates it in the air. High on bright afternoons, near zero at night.
Temperature inversion	Normally air gets colder as you go up. Sometimes it gets warmer instead, and that warm layer acts as a lid: nothing below can rise through it. <i>This is why Delhi's winter nights are so bad.</i>
Mixing depth / boundary layer	How tall the layer of air is that pollution can actually spread into. Two kilometres on a sunny afternoon, sometimes under 100 metres before dawn.
Ventilation coefficient	Mixing depth multiplied by wind speed. How much air per second is available to carry pollution away. <i>Our own calculated number, and one of the model's most useful inputs.</i>
Aerosol	Any tiny particle suspended in air. Smoke, dust, soot. <i>Same thing as particulate matter, different word.</i>
Aerosol optical depth (AOD)	A measure of how much the air blocks light, looking straight up. High AOD means thick haze.
Photolysis	A chemical reaction driven by light. Sunlight splitting a molecule apart. <i>This is how ozone gets made, which is why haze reduces ozone.</i>
Feedback loop / coupling	When an effect circles back and changes its own cause. Pollution reduces sunlight, which increases pollution.
Plume	A trail of smoke travelling away from its source.
Fumigation	When smoke that has been sitting above the city overnight gets mixed down to ground level in the morning. <i>The cause of Delhi's 7 to 10 am pollution spike.</i>
Persistence	The laziest forecast: "tomorrow equals today". The baseline every real forecast must beat.
Held out / hold out	Data deliberately hidden from the model during training, kept back to test it honestly afterwards.
RMSE	Root mean square error. The average size of the model's mistakes. <i>Lower is better.</i>

Gradient boosting	The AI method we use. Hundreds of simple yes/no flowcharts in a row, each correcting the mistakes of the ones before it.
Feature	One input number given to the model. We use 142 of them.
Quantile	Instead of one prediction, a range. "There is a 10 percent chance it is below X and a 90 percent chance it is below Y." <i>How we produce probabilities for GRAP.</i>
GRAP	Graded Response Action Plan. India's official emergency ladder for Delhi air. Restrictions kick in at AQI 200, 300 and 400.
CAMS	The European weather service's global model that includes atmospheric chemistry. We use its output as a starting point.
WRF-Chem	A full research grade weather plus chemistry model. Too heavy to run on our hardware, as explained in section 7.
MoES DSS	The Ministry's own operational forecast system. We use its published numbers as a reference to check ours against. <i>We cite it, we never copy or redistribute it.</i>
CPCB	Central Pollution Control Board. Runs India's official monitoring stations and defines the AQI formula.
FIRMS	NASA's satellite fire detection service. Tells us where stubble is burning.
Snapshot	A saved copy of a full forecast. Our website serves one of these, refreshed every six hours, because the free server cannot run the full calculation itself.